Dasari Manikanta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	6.38
Narayana Junior College	Class XII (Telangana State Board Of Intermediate Education)	2023	98.30%
Auxilium High School	Class X (Board Of Secondary Education)	2021	10.00%

POSITIONS OF RESPONSIBILITY

Core member, Transport DC.

SKILLS

- **Programming Languages:** Python, C, MATLAB, Verilog, LaTeX
- **Libraries and Frameworks:** NumPy, Pandas, Matplotlib
- **Machine Learning:** Supervised Learning, Deep Learning, Model Training and Evaluation
- **Tools and Platforms:** Git, GitHub, Linux (Ubuntu), Google Colab, Cadence, LTspice
- **Embedded Systems and Circuit Design:** Arduino Uno, LTspice, Quartus Prime, DE10-SoC FPGA Board
- **Inter-personal:** Problem Solving, Fast Decision Making, Communication, Leadership

RELEVANT COURSES

- **Mathematics:** Mathematics for Electrical Engineers (Linear Algebra, Calculus), Matrix Theory, Differential Equations and Transform Techniques, Probability, Random Processes
- **Computer Science and AI:** Introduction to Programming (C), Data Structures & Applications, Scientific Programming for Electrical Engineers, Machine Learning
- **Electrical and Electronics:** Circuits and Network Analysis, Engineering Electromagnetics, Digital Systems, Signal Processing, Semiconductor Device Fundamentals, Electrical Machines, Analog Electronics, Power System Analysis, Control Systems **Electrical Labs:** Electrical Circuits Lab, Digital Systems Lab, Digital Signal Processing Lab
- **Other:** Introduction to Life Sciences, English Communication

PROJECTS

Solar PV Inverter Performance & Anomaly Detection | *Python, Pandas, NumPy, Scikit-learn, Time-Series Analysis*

- Developed a data-driven pipeline to analyze **solar PV inverter performance** using operational and weather time-series data.
- Built **expected-generation baselines** and compared them with actual PV output to identify abnormal performance.
- Implemented data-quality checks and **anomaly detection** to identify potential inverter faults, sensor errors, outages, and missing-data periods.
- Analyzed deviations in inverter power output and **quantified associated energy losses** to support performance diagnostics.
- Visualized inverter performance, anomalies, and generation trends using Python-based analytics and plots.

3-Bit Flash ADC Design and Implementation Technologies: Analog Electronics, Comparator Networks, Digital Logic

- Designed and implemented a 3-bit Flash Analog-to-Digital Converter using a parallel comparator architecture.
- Developed the resistor ladder network, comparator stage, and digital encoding logic.
- Verified converter operation for multiple analog input levels using laboratory measurements.
- Analyzed sampling, quantization, encoding, and conversion accuracy through mathematical modeling and experimental validation.

Microphone Amplifier with Class AB Driver Technologies: Analog Electronics, Transistor Biasing, Audio Circuit Design

- Designed and built a low-noise microphone pre-amplifier coupled to a Class AB power driver stage to drive a small speaker.
- Tested the complete amplification chain for sound quality, distortion, and efficiency.
- Measured Total Harmonic Distortion (THD), output noise, and Power Supply Rejection Ratio (PSRR) to characterize signal fidelity.
- Evaluated quiescent current (IQ), gain, bandwidth, and output voltage swing to verify bias stability and power performance.
- Achieved a functional audio amplification system with stable biasing and acceptable distortion levels.

8-Point Radix-2 FFT Processor on FPGA Technologies: Verilog HDL, MATLAB, Intel Quartus Prime

- Designed and implemented a pipelined streaming 8-point Radix-2 FFT processor in Verilog HDL.
- Developed a streaming architecture accepting one complex input sample per clock cycle.
- Verified functional correctness by comparing hardware outputs with MATLAB implementation and Intel FFT IP.
- Evaluated throughput, maximum operating frequency (Fmax), and hardware resource utilization.
- Optimized the pipelined architecture to achieve performance comparable to the vendor-provided FFT IP core.

Eigenvalue Computation using Hybrid Newton-Power Method Technologies: C

- Designed and implemented a hybrid numerical algorithm combining the Power Method and Newton's Method for eigenvalue computation.
- Developed the complete implementation in C without using external numerical or linear algebra libraries.
- Implemented matrix operations, convergence criteria, and iterative optimization from first principles.
- Evaluated numerical accuracy and convergence characteristics on multiple test matrices.